Jianfeng Lin

Yau Mathematical Sciences Center
Tsinghua University
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Office: JingZhai 309
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Research direction: Gauge theory, Floer homology and low dimensional topology.

Personal Data

Education History

University of California, Los Angeles, 09/2012 - 06/2016.
Ph.D. in Mathematics.
Thesis: The Unfolded Seiberg-Witten Floer spectrum: Definition, Property and Applications.
Advisor: Ciprian Manolescu

Peking University, 09/2009-06/2012.
M.A. in Mathematics
Thesis: Fixed subgroups of automorphisms of hyperbolic 3-manifold groups
Advisor: Shicheng Wang

Peking University, 09/2005-06/2009.
B.S. in Mathematics.

Employment History

Yau Mathematical Sciences Center, Tsinghua University, Beijing, China, 09/2021-
Associate professor in Mathematics

University of California, San Diego, La Jolla, US, 07/2019 - 07/2021
Assistant professor in Mathematics

Massachusetts Institute of Technology, Cambridge, US, 09/2016 - 06/2019
C.L.E. Moore instructor in Mathematics
Mentor: Tomasz Mrowka

Publications

Preprints

1. **Exotic diffeomorphisms on closed aspherical 4-manifolds** (with Hokuto Konno, Anubhav Mukherjee and Juan Munoz-Echániz), Submitted (2025) https://arxiv.org/abs/2610.21083
2. **The Dax isomorphism for topological 4-manifolds** (with Yi Xie and Boyu Zhang), Submitted (2026), https://arxiv.org/abs/2609.15792
3. **Non-Existence of Smooth Full-Holonomy Cayley Fibrations** (with Viktor Majewski and Jacek Rzemieniecki), Submitted (2026), https://arxiv.org/abs/2603.26920
4. **Splitting spheres for S^2-links in S^4** (with Yi Xie and Boyu Zhang), Submitted (2026), https://arxiv.org/abs/2608.02785
5. **Non-isotopic surfaces in $T^4\#(S^2 \times S^2)$: an example** (with Yue Wu), Submitted (2026) https://arxiv.org/abs/2604.05805
6. **Pseudo-isotopies of 3-manifolds with infinite fundamental groups** (With Yi Xie and Boyu Zhang), Submitted (2026), https://arxiv.org/abs/2602.09454
7. **Flexible exponents of non-geometric 3-manifolds** (with Zhongzi Wang and Hongbin Sun), Sumitted (2026), https://arxiv.org/abs/2604.23965
8. **Flexible exponent of geometric 3-manifolds and Legendrian maps of Seifert spaces** (with Jianru Duan, Shicheng Wang, Zhongzi Wang and Dongyi Wei), Submitted (2026) https://arxiv.org/abs/2605.17257
9. **Constraints on Lefschetz fibrations with four-dimensional fibers from Seiberg-Witten theory** (with Hokuto Konno, Anubhav Mukherjee and Juan Munoz-Echániz), Submitted (2025) https://arxiv.org/abs/2511.00601
10. **Dax invariants, light bulbs, and isotopies of symplectic structures** (with Weiwei Wu, Yi Xie and Boyu Zhang, Submitted (2025) https://arxiv.org/abs/2501.16083
11. **The monodromy diffeomorphism of weighted singularities and Seiberg–Witten theory** (with Hokuto Konno, Anubhav Mukherjee and Juan Munoz-Echániz), Submitted (2024) https://arxiv.org/abs/2411.12202
12. **On the mapping class groups of 4-manifolds with 1-handles** (With Yi Xie and Boyu Zhang), Sumitted (2024) https://arxiv.org/abs/2501.11821
13. **π_1-injective bounding and application to 3- and 4-manifolds** (with Zhongzi Wang), Submitted (2024) https://arxiv.org/abs/2506.08847.

14. **Configuration space integrals and formal smooth structures** (With Yi Xie), Submitted (2023) `https://arxiv.org/pdf/2310.14156`

Accepted or published

15. **The Family Seiberg-Witten Invariant and nonsymplectic loops of diffeomorphisms** , Accepted: Geometry & Topology.
16. **Homological instability for moduli spaces of smooth 4-manifolds** (with Hokuto Konno), Accepted: Geometry & Topology.
17. **On four-dimensional Dehn twists and Milnor fibrations** (with Hokuto Konno, Anubhav Mukherjee, and Juan Munoz-Echaniz), Accepted: Duke Mathematical Journal.
18. **Family Bauer–Furuta invariant, ExoticSurfaces and Smale conjecture** (with Anubhav Mukherjee). Journal of the association for mathematical research, 3(2):237–275.
19. **Isotopy of the Dehn twist on K3#K3 after a single stabilization.** Geometry & Topology, 27.5 (2023): 1987-2012.
20. **On the Frøyshov invariant and monopole Lefschetz number** (with Daniel Ruberman and Nikolai Saveliev). Journal of Differential Geometry, 123.3 (2023): 523-593.
21. **The unfolded Seiberg-Witten-Floer spectra II: Relative invaraint and the gluing theorem** (with Tirasan Khandhawit and Hirofumi Sasahira). Journal of Differential Geometry, 124(2023), no.2, 231–316.
22. **Intersection forms of spin 4-manifolds and the Pin(2)-equivariant Mahowald invaraint** (with Michael Hopkins, XiaoLin Danny Shi and Zhouli Xu). Communications of the American Mathematical Society, 2 (2022) 22–132.
23. **On the monopole Lefschetz number of finite order diffeomorphisms.** (with Daniel Ruberman and Nikolai Saveliev). Geometry & Topology, 25 (2021) 3591–3628.
24. **A splitting theorem for the Seiberg-Witten invariant of a homology** $S^1 \times S^3$ (with Daniel Ruberman and Nikolai Saveliev). Geometry & Topology 22, no. 5 (2018): 2865-2942.
25. **The Seiberg-Witten equations on end-periodic manifolds and an obstruction to the positive scalar curvature metrics**. Journal of Topology Volume12, Issue2 (2019): 328-371.
26. **The unfolded Seiberg-Witten-Floer spectra I: definition, invariance and examples** (with Tirasan Khandhawit and Hirofumi Sasahira). Geometry & Topology 22, no. 4 (2018): 2027-2114.
27. **SU(2)-Cyclic Surgeries on Knots**. International Mathematics Research Notices, No. 19 (2016), 6018 - 6033.

28. **The A-polynomial and Holonomy Perturbations**. Mathematical Research Letters, Vol. 22, No. 5 (2015), 1401-1416.

29. **Pin(2)-equivariant KO-theory and intersection forms of spin four-manifolds**. Algebraic & Geometric Topology 15 (2015) 863-902

30. **Fixed subgroups of automorphisms of hyperbolic 3-manifold groups** (with Shicheng Wang). Topology Appl., 173 (2014), 209-218.

Selected Awards and Honors

International Congress of Chinese Mathematicians Best Paper Award (2023)

Invited speaker at *Current Developments in Mathematics* (2025)

Service

Member of the editorial boards of Geometry & Topology

Referee for 10 mathematical journals

Supervision: 3 postdoctoral researchers, 8 Ph.D. students